

Three zero-length wormholes on a circle: the sharp diameter is one quarter

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Abstract

Identifying at most three pairs of points on a circle of circumference L cannot reduce its continuous path-metric diameter below $L/4$. Two antipodal identifications attain this value, so the optimum for three zero-length wormholes equals that for two. The statement includes repeated, shared and trivial endpoint pairs. We reduce the problem to connected degree-four multigraphs on three vertices and prove that their total edge length is at most four times their diameter. Four graph types suffice; the only loop-free type is handled by two elementary linear identities. An accompanying exact certificate cover has 151 closed cells over the 15 pairings of six endpoint occurrences. The proof holds for all real nonnegative arc lengths and does not depend on finite testing.

1 Statement and context

Let $S_L = \mathbb{R}/L\mathbb{Z}$, where $L > 0$, carry its intrinsic circle metric. A *zero-length wormhole* is an unordered pair of points that may be traversed at zero cost. Given a list $\mathcal{W}$ of such pairs, let $X(\mathcal{W})$ be the metric quotient obtained by allowing those jumps and taking the induced path metric. Distances and diameter refer to all points, including points in the interiors of the remaining arcs. Define

$$D_m(L) = \inf_{|\mathcal{W}| \leq m} \text{diam } X(\mathcal{W}), \quad D_m = D_m(1).$$

Pairs may share endpoints, be repeated, or be trivial pairs (a, a) .

Theorem 1. *For every $L > 0$ and every list $\mathcal{W}$ of at most three wormholes on S_L , one has*

$$\text{diam } X(\mathcal{W}) \geq \frac{L}{4}.$$

Equality is attained by identifying 0 with $L/2$ and $L/4$ with $3L/4$. Consequently $D_3(L) = D_2(L) = L/4$, and in particular $D_3 = 1/4$.

Ahn, Cabello, Cheong, Garijo and Moon [1] relate the length of a connected metric graph to its diameter, cyclomatic number and number of leaves. Their Theorem 2 gives a lower bound of $L/5$ in the three-wormhole problem. In their circle application they give the two-wormhole value $L/4$ and conjecture that $2n + 1$ wormholes have the same optimum as $2n$ wormholes. Theorem 1 proves the three-wormhole layer by improving $L/5$ to $L/4$. No assertion about higher odd budgets or a classification of all equality configurations is made here.

The lower bound below is self-contained. Its main ingredient is the following finite-graph statement. A loop contributes two to the degree; parallel edges are allowed. With nonnegative

lengths, edges of length zero are contracted in the metric, while total length means the sum of the assigned edge lengths.

Proposition 2. *Every connected multigraph G on three vertices, each of degree four, with nonnegative edge lengths satisfies*

$$\text{len}(G) \leq 4 \text{diam}(G). \quad (1)$$

The use of diameter in graph extremal inequalities also occurs in Zeng's work on induced forests in even-hole-free graphs [2]. That work is adjacent discrete-graph context: its diameter is a vertex distance parameter, whereas the present proof concerns continuous edge interiors. It is not used as a proof ingredient. The literature comparison here is confined to the cited sources and bounded searches; it makes no exhaustive priority claim.

2 Distances and degenerate lengths

Lemma 3. *Let two distinct nonloop edges share an endpoint u , have lengths $\ell, m > 0$, and have other endpoints v, w , respectively. In the ambient metric graph, put $r = d(v, w)$. Then*

$$\max_{x \in uv, y \in uw} d(x, y) = \frac{\ell + m + r}{2}. \quad (2)$$

This includes parallel edges, for which $v = w$ and $r = 0$.

Proof. Write $p = d(u, v)$ and $q = d(u, w)$. Coordinates $s \in [0, \ell]$ and $t \in [0, m]$, both measured from u , give four endpoint-route lengths:

$$s + t, \quad s + m - t + q, \quad \ell - s + t + p, \quad \ell + m - s - t + r. \quad (3)$$

Their minimum is $d(x, y)$. Indeed, each route is a valid walk, even if its shortest vertex-to-vertex portion revisits an edge. Conversely, a shortest path from an interior point of one edge to the other must leave and enter those edges at endpoints. Endpoint cases follow by continuity.

The first and fourth expressions sum to $\ell + m + r$, giving the upper bound. For the reverse bound, take

$$s = \frac{\ell + r - q}{2}, \quad t = \frac{m + q}{2}.$$

These coordinates are feasible: $q \leq p + r \leq \ell + r$, $r \leq p + q \leq \ell + q$, and $0 \leq q \leq m$. The first, second and fourth routes equal $(\ell + m + r)/2$, and the third is larger by $p + q - r \geq 0$. $\square$

Two further observations will supply diameter witnesses. A loop of length h has a midpoint at distance $h/2$ from its attachment vertex. An isometrically embedded cycle of length c has an antipode at distance $c/2$ from any specified vertex on it. If a loop attaches to that cycle at the specified vertex, its midpoint and the antipode have distance $(h + c)/2$. Also, midpoints of loops of lengths h, k at any two vertices are at distance at least $(h + k)/2$.

Lemma 4. *For a fixed connected finite multigraph, its continuous diameter is a continuous function of its nonnegative edge lengths, with zero-length edges contracted in the metric.*

Proof. Vertex distances are minima of the lengths of finitely many simple paths, and hence are continuous in the edge lengths. For two distinct edges, the distance between specified points is the

minimum of the four endpoint routes, as in the proof of Lemma 3. For two points on the same edge of length ℓ and endpoints u, v , separated along the edge by h , their distance is

$$\min\{h, \ell + d(u, v) - h\}.$$

The direct segment and the complementary endpoint route give this formula; routes through one endpoint or in the reverse orientation cannot improve the direct segment. It also holds for loops.

Parameterize points on each edge by fractions in $[0, 1]$ of its length. The displayed distance formulas are jointly continuous in the lengths and the fractional positions, including when an edge collapses. They are uniformly continuous on a compact neighborhood of any fixed length vector times $[0, 1]^2$. Taking maxima over these compact squares and the finitely many edge pairs preserves continuity. $\square$

It therefore suffices to prove Proposition 2 for positive lengths: replace each length $x_i \geq 0$ by $x_i + \varepsilon$, apply the positive-length bound, and let $\varepsilon \downarrow 0$. Both length and diameter converge. This step needs no fixed-total-length normalization.

3 The four graph types

Let n_{ij} count edges between distinct vertices $i, j \in \{0, 1, 2\}$, and let k_i count loops at i . The degree equations are

$$n_{ij} + n_{ik} + 2k_i = 4 \quad (\{i, j, k\} = \{0, 1, 2\}).$$

Connectivity excludes $k_i = 2$ and $n_{ij} = 4$, so $k_i \in \{0, 1\}$ and $n_{ij} \leq 3$. The three cross-edge multiplicities have the same parity. If odd, they are $(1, 1, 1)$ or a permutation of $(3, 1, 1)$: two multiplicities equal to three would violate a degree equation. If even, they are zero or two, and at least two must be nonzero. Thus Table 1 exhausts the possibilities up to relabeling.

Type	Description	Cross-edge multiplicities
I	Triangle with a loop at each vertex	$(1, 1, 1)$
II	Doubled path with a loop at each end	$(2, 2, 0)$
III	Triple bundle, two-edge return path, one loop	$(3, 1, 1)$
IV	Doubled triangle	$(2, 2, 2)$

Table 1: Connected degree-four multigraphs on three vertices.

Write $D = \text{diam}(G)$. We now prove (1) for each type.

Type I. Let the central triangle have total length c and the loops have lengths h_0, h_1, h_2 . The triangle is isometric, since excursions into loops cannot shorten paths between its points. Its antipode to vertex 0 and the midpoint of loop 0 give $2D \geq c + h_0$. The other two loop midpoints give $2D \geq h_1 + h_2$. Adding counts every edge once.

Type II. Let the two parallel-edge cycles have lengths c_1, c_2 , and let the loops at the outer vertices have lengths h_1, h_2 . Both cycles are isometric: every exterior component attaches to a chosen cycle at only one vertex. Pair the midpoint of each end loop with the antipode to its attachment in the adjacent cycle. This gives $2D \geq c_1 + h_1$ and $2D \geq c_2 + h_2$, whose sum proves the bound.

Type III. Let u, v be the triple-bundle endpoints and w the remaining vertex. Write the bundle lengths as x, y, z , choosing x shortest, and let p, q be the lengths on uw, wv , with a loop of length h at w . The cycle consisting of x, p, q is isometric: any exterior $u-v$ edge can be replaced by x at no greater cost, and loop excursions do not help. Its antipode to w and the loop midpoint give $2D \geq x + p + q + h$. Lemma 3 applied to the remaining two parallel edges gives $2D \geq y + z$. Adding again proves the bound. Ties in the choice of the shortest edge are harmless.

Type IV. For the bundles on UV, VW, WU , respectively, write their lengths as

$$a \leq A, \quad b \leq B, \quad c \leq C.$$

Every shortest vertex path can use the shorter edge of each bundle. A simple path is direct or passes through the third vertex, so

$$d(U, V) = \min(a, b + c), \quad d(V, W) = \min(b, a + c), \quad d(W, U) = \min(c, a + b).$$

Set $S = A + B + C + a + b + c$. If a, b, c satisfy the triangle inequalities, Lemma 3 applied to adjacent pairs of longer edges gives the nonnegative slacks

$$q_1 = 2D - A - B - c, \quad q_2 = 2D - B - C - a, \quad q_3 = 2D - C - A - b.$$

Direct expansion gives the identity

$$4D - S = \frac{2}{3}(q_1 + q_2 + q_3) + \frac{1}{3}((A - a) + (B - b) + (C - c)) \geq 0. \quad (4)$$

In the remaining regions one short side dominates the sum of the other two. Suppose $a \geq b + c$. Then $d(V, W) = b$ and $d(W, U) = c$, so q_1 and q_3 remain nonnegative. Here the needed identity is

$$4D - S = q_1 + q_3 + (A - a) \geq 0. \quad (5)$$

Permuting the bundles treats $b \geq a + c$ and $c \geq a + b$. The triangle region and these three weak-dominance regions are closed and cover all nonnegative triples, including their common walls. This finishes the positive-length proof of Proposition 2; Lemma 4 supplies every zero-length face.

4 From the circle to the graph, and sharpness

Pad a list of fewer than three wormholes with trivial pairs. Sort the six endpoint *occurrences* cyclically, breaking ties arbitrarily, and retain the six consecutive arc lengths $x_0, \dots, x_5 \geq 0$. Their sum is L . If all occurrences coincide, the wraparound arc still carries the full length L .

Identify each prescribed pair of occurrences. Before contracting zero edges this produces three labeled vertices, each with four incident half-edges. The original cyclic chain makes the graph connected. Zero-length arcs identify coincident occurrences; the prescribed pairings perform the wormhole identifications. Together, including their transitive closure, these are exactly the zero-cost identifications in $X(\mathcal{W})$. Its metric is therefore the resulting graph path metric, with zero edges contracted. This accounts for shared, repeated and trivial pairs without requiring six distinct physical points. Proposition 2 now gives $L \leq 4 \operatorname{diam} X(\mathcal{W})$.

For sharpness, identify $0 \sim L/2$ and $L/4 \sim 3L/4$. The quotient consists of four parallel edges of length $L/4$. By Lemma 3, every distinct-edge maximum is $L/4$. Distances on one edge are at most its length $L/4$. Thus the full diameter is exactly $L/4$, proving Theorem 1.

5 Exact certificates and computational disclosure

The argument above proves the universal statement. The supplementary certificate data record it as exact rational inequalities. There are $5 \cdot 3 \cdot 1 = 15$ pairings of six cyclically labeled occurrences: pair the least unused label with any other unused label and recurse. These give 2, 3, 6, 4 instances of Types I–IV, respectively. For Types I and II one cell suffices. Type III uses the three choices of a shortest bundle edge. Type IV uses the 2^3 choices of shorter edges and the four closed short-side regions. The total is

$$2 \cdot 1 + 3 \cdot 1 + 6 \cdot 3 + 4 \cdot (8 \cdot 4) = 151.$$

Weak inequalities ensure coverage of ties and degenerate faces. Each cell records nonnegative inequalities whose nonnegative rational linear combination is $4D - \sum_{i=0}^5 x_i$.

A separately implemented checker reconstructed the geometric meaning of all 1,852 rows and verified all 151 coefficient identities using exact rational arithmetic. A further check used integer breadth-first search on half-unit subdivisions, after contracting zero edges, to obtain exact distances between sampled points. These distances give lower bounds on the continuous diameter. The test covered all 792 nonnegative integer compositions of 7 into six parts for each of the 15 pairings, together with five probes per pairing, for 11,955 configurations. The probes include the all-zero vector as an additional control outside the positive-circumference theorem. All tests passed; all 151 cells were exercised. These finite checks supplement the geometric proof and do not establish its real-length quantifiers.

The accompanying reproducibility instructions identify the certificate data and the two verification implementations. AI assistance was used to develop the argument, generate verification code, draft the text, and conduct a separate model-context mathematical review. That review is not external peer review or a claim of statistical independence. No proof-assistant formalization is claimed.

References

- [1] Hee-Kap Ahn, Sergio Cabello, Otfried Cheong, Delia Garijo and Jeongwon Moon, *Diameter and Length of Metric Graphs*, arXiv:2608.08716v1 (2026). <https://arxiv.org/abs/2608.08716>.
- [2] Zijian Zeng, *A sharp diameter bound for induced forests in even-hole-free graphs*, Zenodo preprint (2026). <https://doi.org/10.5281/zenodo.22294612>.